

Voice Agents Cheat Sheet

Baseline written 2026-10-01: LangChain 1.4.3, FastAPI 0.142.2 (Python 3.12), Spring AI 2.0.1 on Spring Boot 4.1.x (Java 21), Pipecat 1.12.0, LiveKit Agents 1.8.3 (server 1.13.7), OpenAI gpt-live-1 / gpt-realtime-2.1, Gemini gemini-3.8-live, elevenlabs 2.70.0, faster-whisper 1.2.1, whisper.cpp 1.9.4, Silero VAD 6.2.3, Asterisk 23.5 / 22.11 / 20.21, twilio 9.11.2. Snippets are illustrative, not compiled; model names change, check the vendor models page.

ARCHITECTURES

- Cascaded** STT → LLM → TTS: reuse the text core (ask(conversationId,userId,text)), any model, on-prem possible.
- Speech-to-speech** (GPT-Live, Realtime, Gemini Live, Voice Live): lowest latency, natural prosody, one vendor model; core via tool call / MCP.
- Hybrid**: vendor conversation layer, your answers (GPT-Live delegation, ElevenLabs custom LLM).
- RAG sits in the LLM step or a tool; stream every stage; chunk TTS by sentence; write answers for the ear (no markdown).

LATENCY BUDGET

T_resp ≈ T_net + T_eot + T_TTFT(LLM) + T_TTFB(TTS)
Speech-to-speech: T_net + T_eot + T_model.
Terms add; fix the largest first. Retrieval is inside TTFT: filler, cache, small reranker. Record t_user_end, t_stt_final, t_llm_first_token, t_tts_first_byte, t_audio_sent; report p50/p95/p99.

TURN-TAKING CHECKLIST

- VAD (Silero, TEN VAD) in front of Whisper-type STT to avoid hallucinated text.
- Endpointing: silence (300-800 ms) < STT events (Deepgram Flux EndOfTurn/EagerEndOfTurn, AssemblyAI end_of_turn) < semantic (LiveKit turn detector, Pipecat Smart Turn).
- Barge-in: cancel LLM + TTS, flush queued audio (Twilio clear, Asterisk FLUSH_MEDIA), truncate history to what was heard; stop within ~300 ms.
- Echo: browser AEC/NS on in getUserMedia; headset test; phone lines echo less.
- DTMF, silence timeouts, backchannels; replay 50 consented calls when tuning.

SPEECH TO TEXT

Option	Note
Deepgram Nova / Flux	streaming, end-of-turn events, keyterms
AssemblyAI streaming	turn messages, formatting
OpenAI gpt-4o-transcribe, Azure Speech	REST + Realtime transcription; phrase lists
faster-whisper (MIT)	CTranslate2, CPU/GPU, windowed
whisper.cpp (MIT)	C/C++, Apple silicon, small devices
Moonshine (MIT)	small, on-device, short utterances
Parakeet / Kyutai STT (CC-BY-4.0)	GPU; attribution required

WER=(S+D+I)/N. 8 kHz telephone audio hurts; send native rate, set language, add vocabulary, sample WER per channel.

TEXT TO SPEECH

Model	Licence
Kokoro-82M	Apache-2.0; 24 kHz; default local
Piper (piper1-gpl)	GPL-3.0 (original repo archived); voices have own terms
XTTS-v2 (idiap fork)	code MPL-2.0; weights non-commercial
Sesame CSM-1B	Apache-2.0; speech only, GPU
Kyutai TTS	CC-BY-4.0; streaming text in, GPU

Hosted: ElevenLabs, OpenAI, Cartesia, Azure.
Stream chunks; SSML where supported. Cloning needs the speaker's consent; disclose the AI voice.

AUDIO FORMATS

Format	Where
mu-law 8 kHz 8-bit	Twilio Media Streams, US PSTN (base64, 20 ms)
A-law 8 kHz	EU PSTN, SIP trunks
PCM16 16 kHz	most STT, slin16, ElevenLabs pcm_16000
PCM16 24 kHz	Kokoro, OpenAI PCM
Opus 48 kHz	WebRTC (transport decodes)

24 kHz → 8 kHz: polyphase resample_poly(x,1,3) then lin2ulaw; keep resampler state across chunks.

REALTIME API TRANSPORTS

API	Start / connect
OpenAI GPT-Live gpt-live-1	POST /v1/live/sessions; WebRTC, WebSocket, SIP; delegate to your backend
OpenAI Realtime	POST /v1/realtime/client_secrets (ephemeral) or /v1/realtime/calls; WebRTC, WS, SIP; function tools, remote MCP
Gemini Live	WebSocket, function calling, ephemeral tokens
Azure Voice Live	STT + model + TTS behind one API; semantic end-of-turn

Never ship the API key to the browser: mint ephemeral secrets in FastAPI/Spring; cap session length.

ELEVENLABS CUSTOM LLM

- Agent → LLM → Custom LLM: URL of an OpenAI-compatible /v1/chat/completions (or Responses) endpoint, secret as Bearer.
- Answer as SSE: data: {chunk} ... data: [DONE]; last user message = transcript.
- Identity: elevenlabs_extra_body (user id, conversation id) → ask().
- Docs under /docs/eleven-agents/; Twilio native + SIP; web widget over WebRTC.

PIPECAT SKELETON

```
pipeline = Pipeline([transport.input(), st
t,
user_agg, llm, tts,
transport.output(), assistant_agg])
worker = PipelineWorker(pipeline,
params=PipelineParams(enable_metrics=Tru
e))
# python bot.py -t webrtc | daily | twilio
```

Frames flow through processors; VAD on LLMUserAggregatorParams; transports SmallWebRTC, WebSocket, Daily. Smart Turn; Flows in core (pipecat.flows).

LIVEKIT AGENTS SKELETON

```
server = AgentServer()
@server.rtc_session(agent_name="docs")
async def entrypoint(ctx):
    s = AgentSession(stt=..., llm=..., tts
=...,
turn_handling=TurnHandlingOptions(
turn_detection=inference.TurnDetector
()))
    await s.start(room=ctx.room, agent=A())
```

Pipeline or realtime model; python agent.py dev|start; LiveKit SIP for phones; self-host or Cloud.

TWILIO CONVERSATIONRELAY (TEXT WS)

Dir	type
in	setup, prompt{voicePrompt,last}, interrupt{utteranceUntilInterrupt}, dtmf, error
out	text{token,last}, play, sendDigits, language, end{handoffData}

TwimML <Connect><ConversationRelay url="wss://...">. Twilio does STT, endpointing, TTS. Validate X-Twilio-Signature.

TWILIO MEDIA STREAMS (AUDIO WS)

Dir	event
in	connected, start{streamSid}, media{payload}, dtmf, mark, stop
out	media{payload}, mark, clear (barge-in)

<Connect><Stream url/> bidirectional; 8 kHz mu-law base64. Use mark to know what was played.

ASTERISK: ARI EXTERNAL MEDIA

```
Stasis(docs=assistant,7000) ; dialplan
POST /ari/channels/externalMedia?app=docs-a
ssistant
&transport=websocket&encapsulation=none
&external_host=docs-voice-service&format=s
lin16
AudioSocket(${UUID},voice:9019) ; 0x01 uu
id 0x10 pcm 0x03 dtmf
```

- RTP (UDP) < AudioSocket (TCP, 3-byte header) < chan_websocket (23.0+/22.6+/21.11+/20.16+): binary audio, JSON text control MEDIA_START, FLUSH_MEDIA, MARK_MEDIA, MEDIA_XOFF/XON; queue ~1000 frames.
- Also FreeSWITCH mod_audio_stream, Jambonz, SIP to OpenAI / ElevenLabs / LiveKit. TLS + SRTP, anti-toll-fraud.

SPRING AI 2.0 AUDIO

```
TranscriptionModel.transcribe(resource)
TextToSpeechModel.call(text) ; .stream(tex
t)
// Flux<byte[]>; StreamingTextToSpeechModel
1.0 SpeechModel is gone.
spring.ai.model.audio.speech=openai|elevenlabs.
Spring WebSocket handler + VAD + ChatClient for
a hand-rolled pipeline; cancel subscriptions on
barge-in.
```

PRODUCTION CHECKLIST

- c=λ⋅T_call; admit or reject at a measured per-pod limit.
- GPU: RTF = compute / audio seconds; load models once.
- c_min = c_tel + c_stt f_u + c_tts f_a + c_llm + c_infra; cap idle time.
- AI disclosure + per-call recording consent; set retention.
- Redact PII before logs and traces (Presidio).
- Per-stage latency, interruption rate, sampled WER.
- Test: recorded calls, simulated callers, barge-in, chaos.
- Backup providers; human transfer; non-voice path.